

LECTURE L11-12 • MODULE 2 • CO2

Multivariate structure

Week 6 | Slides: DATA209_Module_2_Core_Advanced_EDA.pptx, pages 21-29 | Laboratory: P11-12

TOPICS COVERED

- Multicollinearity and the variance inflation factor
- Identifying groups in data
- K-means for structure discovery
- Interpreting clusters

Why go past two variables

A relationship that holds within every subgroup can **reverse** when the subgroups are pooled. That is Simpson's paradox, and it is the strongest argument for multivariate analysis: two variables at a time can be consistently, confidently wrong.

Multicollinearity

Multicollinearity is strong correlation *among the predictors themselves*, not with the target. Its consequences:

- Coefficient estimates become unstable — small data changes swing them, sometimes flipping sign.
- Standard errors inflate, so genuinely useful variables appear insignificant.
- Predictive accuracy may be unharmed; **interpretability is destroyed**.

It is invisible in a target-correlation ranking. You must inspect the predictor-to-predictor block of the correlation matrix, then confirm with the **variance inflation factor**.

VIF for a predictor is $1 / (1 - R^2)$ from regressing that predictor on all the others. It answers: how redundant is this column?

VIF	READING	ACTION
≈ 1	uncorrelated with the others	keep
1 – 5	moderate	keep, note it
5 – 10	coefficients becoming unreliable	investigate the pair
> 10	largely reconstructable from the others	drop, combine, or reduce

In the shoppers data, **BounceRates** and **ExitRates** measure nearly the same thing. Keep one, or combine them — a decision revisited in L25-26.

K-means, exploratorily

Four steps: choose k and initialise (k-means++ rather than at random); **assign** each point to its nearest centroid; **update** each centroid to the mean of its members; **repeat** until assignments stop changing.

Scale the features first. K-means minimises Euclidean distance, so an unscaled variable with a large range decides the clusters by itself.

Choosing k , and being honest about the answer

The **elbow** on the inertia curve and the **mean silhouette** are the two standard aids. Silhouette compares each point's distance to its own cluster against the nearest other cluster, running from -1 to $+1$:

SCORE	READING
> 0.7	strong, well-separated
$0.5 - 0.7$	reasonable structure
$0.25 - 0.5$	weak; clusters overlap
< 0.25	no substantial structure

On standardised `pima.csv` the best score across $k = 2 \dots 6$ is about 0.197. The correct conclusion is that this dataset has no strong natural grouping — and reporting that honestly is a good answer. Manufacturing clusters that are not there is not.

Interpreting clusters

A cluster is only real if you can describe it. Profile each one by computing every feature's mean within it and comparing to the overall mean; name it in domain language ("older, higher BMI, high glucose"), not as "cluster 2". Check sizes — a cluster of four points is an outlier group, not a segment. Re-run with a different seed and a different k : structure that vanishes was never there.

KEY TERMS

Simpson's paradox	Multicollinearity	Variance inflation factor	k-means	k-means++
Inertia	Elbow method	Silhouette score	Cluster profiling	

COMMON ERRORS TO AVOID

- Running k-means on unscaled features
- Reporting clusters without profiling or naming them
- Choosing k from the elbow alone without a silhouette check
- Presenting weak clustering as clean market segments

READINGS — ALL FREE TO ACCESS

1. **scikit-learn — clustering**

<https://scikit-learn.org/stable/modules/clustering.html>

2. **statsmodels — variance inflation factor**

https://www.statsmodels.org/stable/generated/statsmodels.stats.outliers_influence.variance_inflation_factor.html

3. **Hastie et al. — Elements of Statistical Learning, Ch. 14**

<https://hastie.su.domains/ElemStatLearn/>

TEST YOURSELF

1. A predictor has VIF 14. Name three possible responses.
2. Why can multicollinearity leave predictive accuracy intact while ruining interpretation?
3. Mean silhouette is 0.19 at every k you tried. What do you report?
4. Give a concrete example of Simpson's paradox.